MandiMitra

Explainable mandi price intelligence

Does AI really beat “same as today” for wholesale onion and tomato prices? We tested it — properly.

Vishnu Keer · Vivekananda Global University, Jaipur

Live app

Code + data

mandimitra.streamlit.app · github.com/vishnu-keer/mandimitra

Price-prediction claims reach farmers unchecked

Rs. 600
→ Rs. 4,500

per quintal: Jaipur wholesale onion, June → 21 October 2013, then back to about Rs. 800 by December

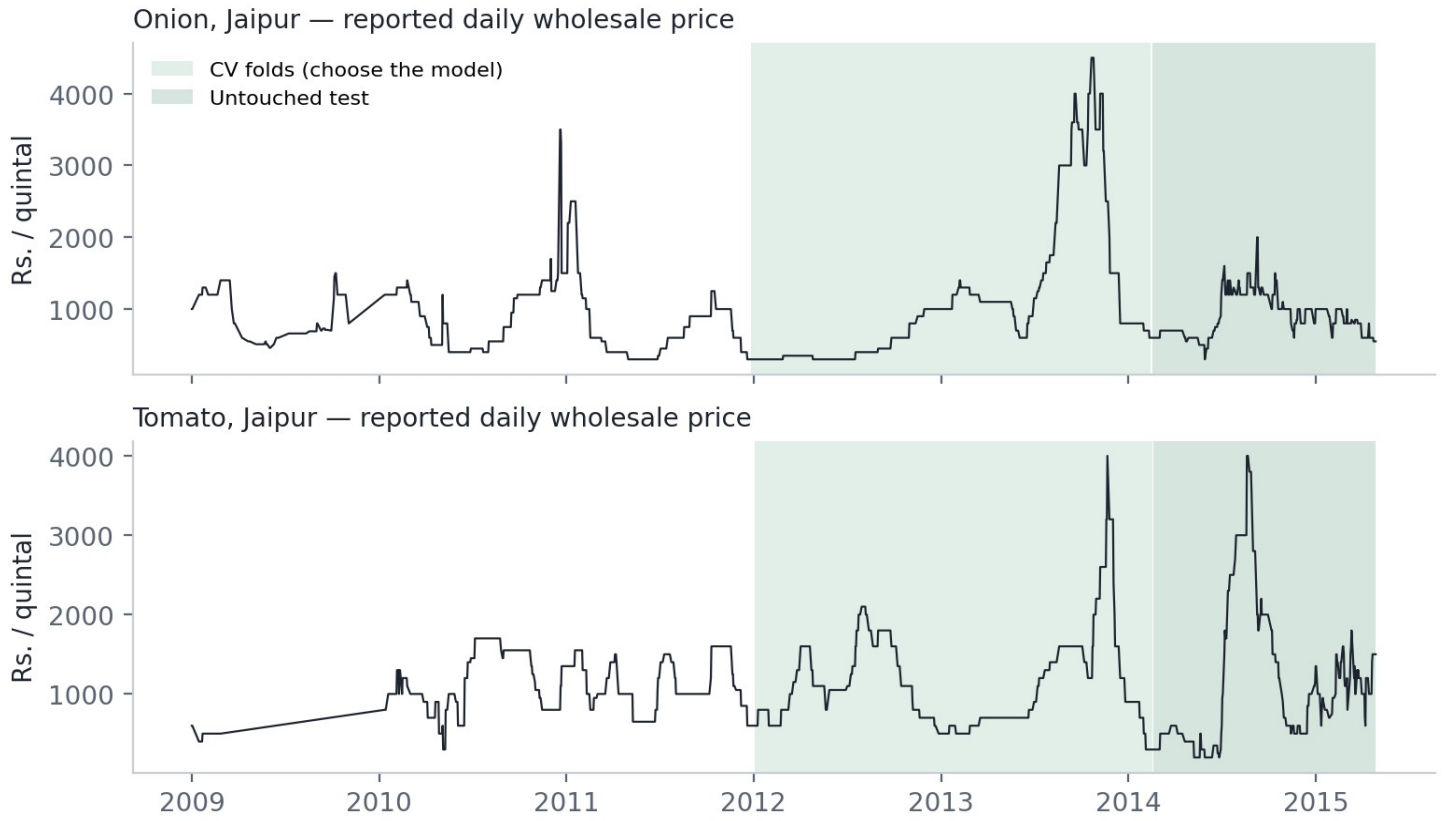
Official Department of Consumer Affairs series

1 **Farmers**
Sell today or hold? A confident wrong forecast costs real money.

2 **Traders**
How noisy is this market, and how stale is the quoted price?

3 **Students & analysts**
Was this model tested in a way that could have shown it fails?

Prices move in rare jumps that price history alone cannot see



84.8%

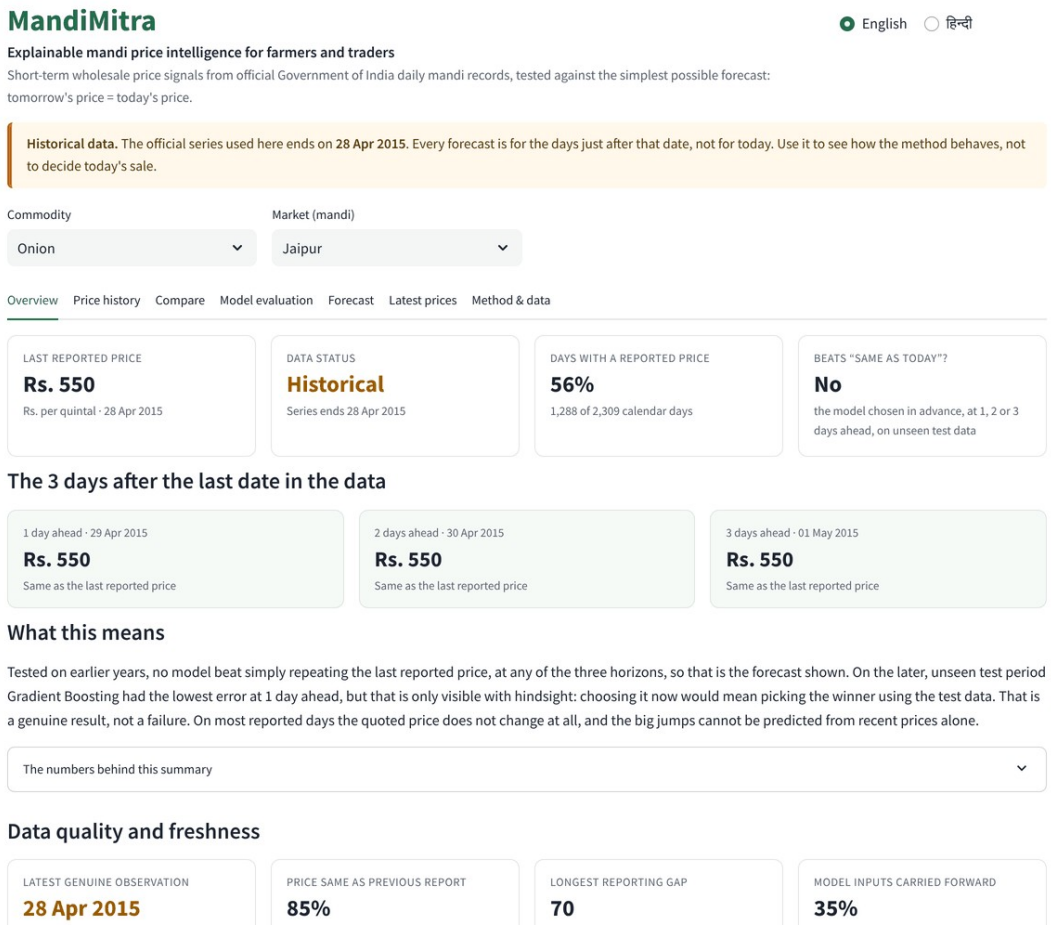
of reported Onion/Jaipur days quote exactly the previous price (Tomato: 82.8%)

~55%

of calendar days have a reported price at all (2009–2015)

Data: Dept. of Consumer Affairs via data.gov.in (GODL-India). Related evidence: onion price volatility is linked to market-arrivals uncertainty (Birthal, Negi & Joshi, 2019, JADEE 9(3)).

An honest dashboard, not a model leaderboard



Strong reference first

Every forecast is shown next to “same as today” and the last reported price.

Freshness and quality

Latest genuine observation, stale-quote rate, reporting gaps — on screen.

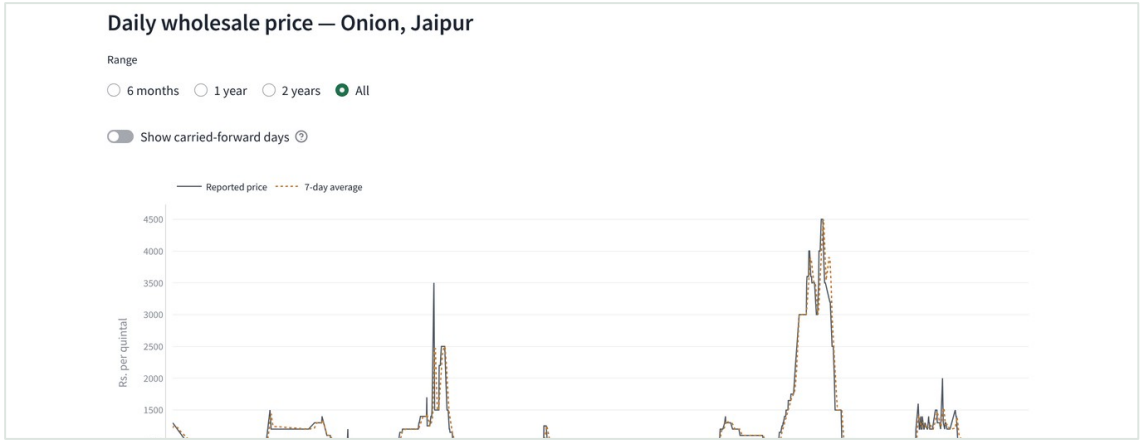
Honest uncertainty

Ranges are drawn only if they held up on unseen data. They didn't, so we say so.

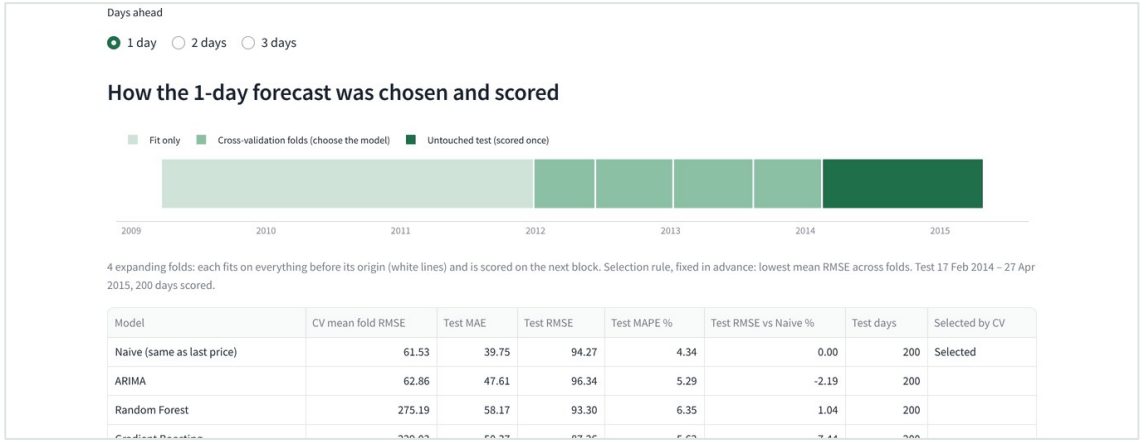
English + हिन्दी

Plain-language views in Hindi; CSV downloads; works on a phone.

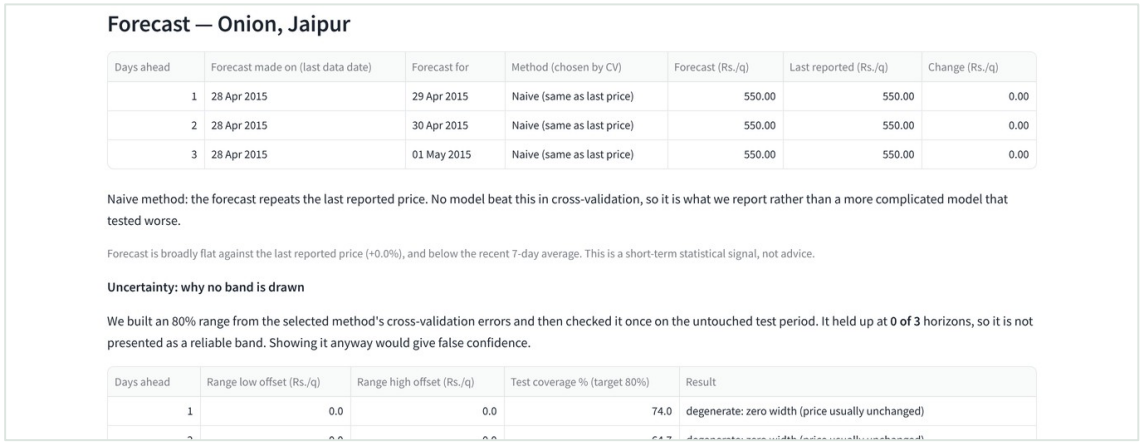
It works today: seven views, every number traceable to a file



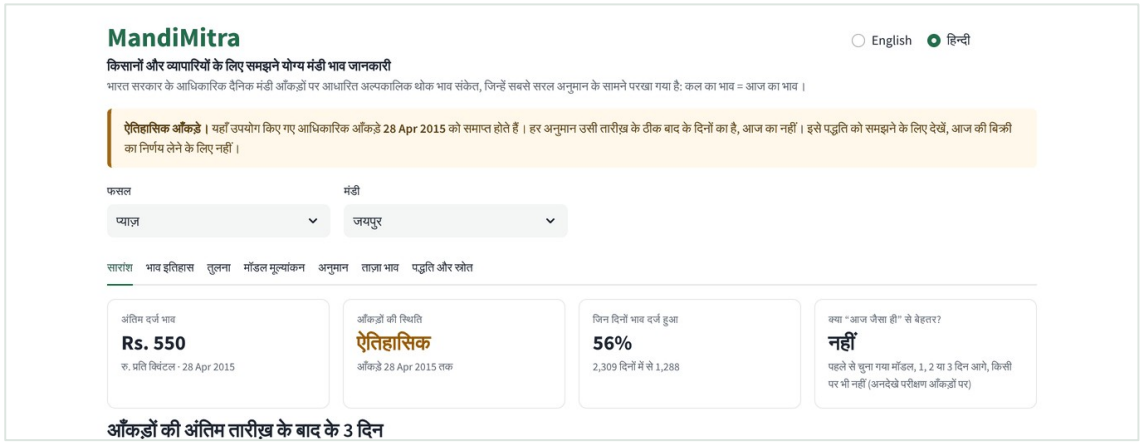
Price history: reported vs carried-forward days



Model evaluation: chosen by CV, every model on the same days

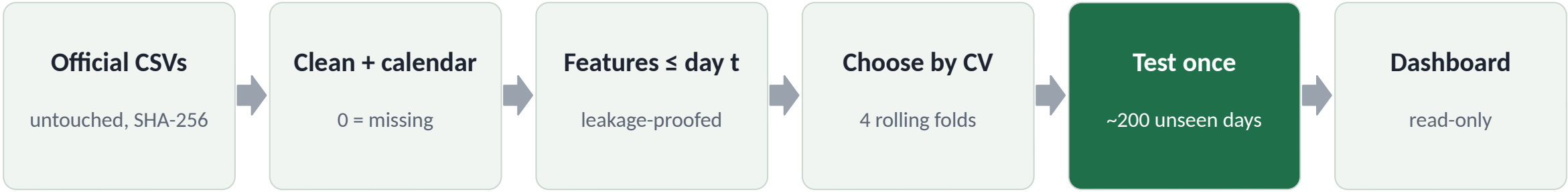


Forecast: origin and target dates, no fake confidence band



Hindi view for farmers

A pipeline that refuses to write a misleading result



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automated tests: leakage, chronology, embargo, identical rows, provenance, ingestion

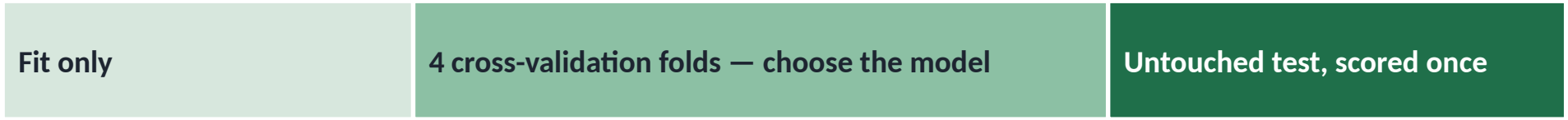
2 gates

pipeline exits with FAIL rather than publish a broken comparison

Byte-identical

rebuild from the government files on Python 3.11 and 3.12; CI on every push

The model is chosen before the test data is ever scored



Naive

tomorrow = today

ARIMA

classical time series

Random Forest

tabular ML

Gradient Boosting

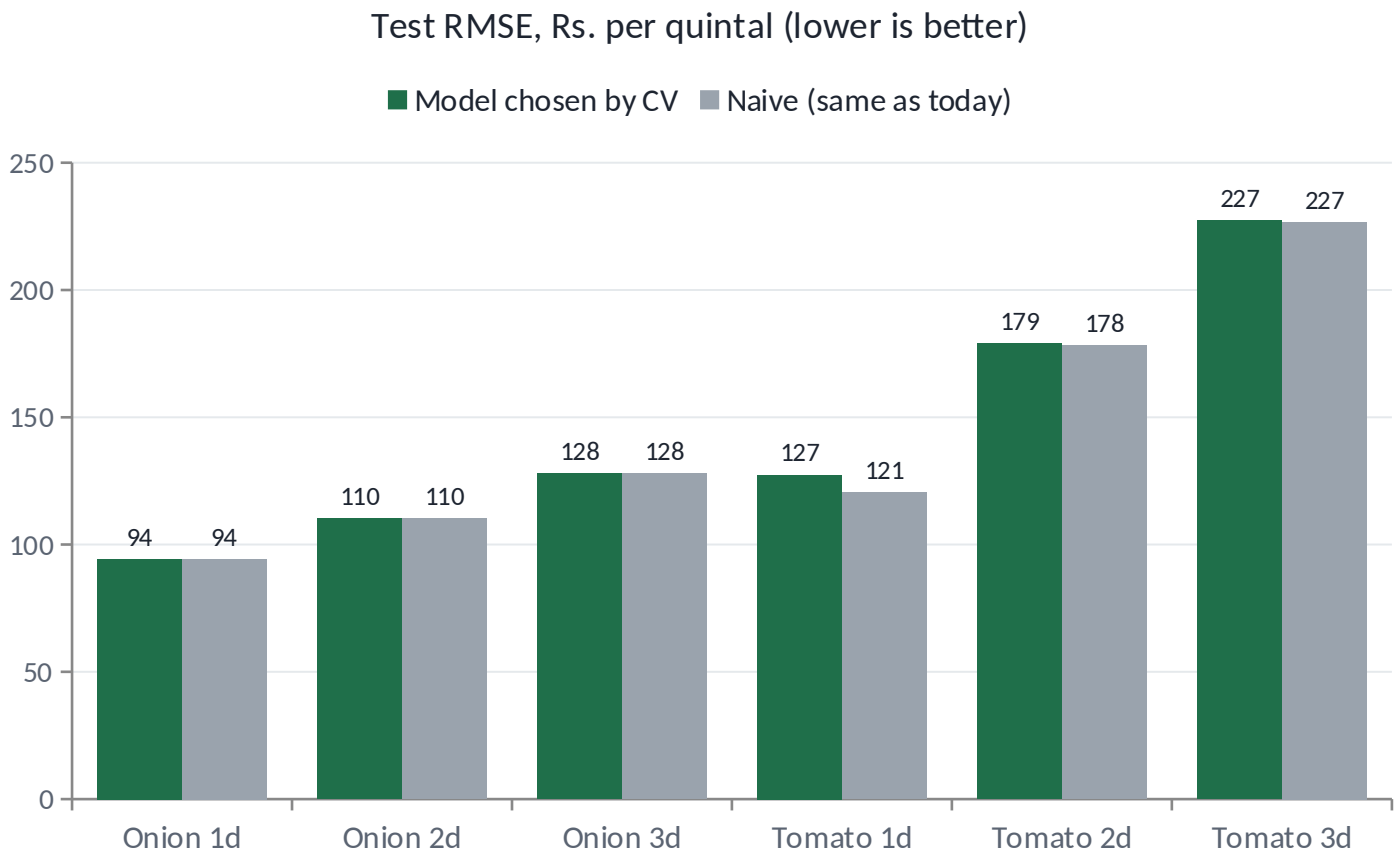
tabular ML

RF-delta

predicts the change

- Selection rule written in advance: lowest mean RMSE across the four folds.
- Every model scored on identical days; only days the mandi really reported are scored.
- Training rows whose target date crosses a boundary are removed (embargo).
- Significance: Diebold–Mariano tests vs Naive, Holm-adjusted.

The model chosen in advance never beat “same as today”



0 of 48

significance tests found any model more accurate than Naive

p = 0.505

for our first version's “+7.33%” — it came from picking the winner on the test set

Naive also had the lowest error on an ordinary day (MAE) at all six.

Onion: CV chose Naive at all horizons. Tomato: 1d RF-delta −5.66%, 2d ARIMA −0.53%, 3d ARIMA −0.31% vs Naive. Files: outputs/best_model.csv, outputs/experiments/dm_test.csv.

No user evidence yet — this is exactly how we will get it

What we can claim today

- A working, tested, reproducible product.
- A measured, significance-tested result on official data.
- No farmer interviews, user counts, savings or income effects — none have been measured.

Next 4 weeks: consented user study

- 5–10 traders and farmers at a Jaipur mandi, with informed consent, no names stored.
- Measure: can they read the forecast, the reference and the warning correctly?
- Measure: which screen would change a decision, and how?
- Publish the anonymised findings — whatever they are.

Rs. 0 to run, open data, and built to add markets

Rs. 0

GitHub + Actions, Streamlit Community Cloud, data.gov.in. No paid service anywhere.

GODL-India

Government open data with attribution; code MIT. Licences kept separate.

Live-ready

Validated client for the official AGMARKNET daily price API — waiting only for a free access key.

Gate-based

New market = add its official file. The 400-row gate decides; nothing is forced.

What this does not show yet — and the next milestones

Limitation	Next milestone
Data ends April 2015	Connect the official daily API (key); check a CEDA daily mirror's terms
One mandi, two commodities	Run more Rajasthan and national centres through the gate
Price history only	Add arrivals and rainfall — where a real edge is more plausible
Back-test only	Score live forecasts against prices published afterwards
No user evidence	Consented study with 5–10 traders and farmers

Trust before prediction

We ask for:

- Mentors in agricultural economics and forecasting
- Introductions to a mandi committee or FPO for a consented pilot
- Feedback on what traders would actually use

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